SANGER: CLONED AMPLICONS

one clone is one molecule, read once



529-bp window that the clones read cleanly;
447 chromatograms from the six mice of Figure 5

● mutation ● sequencing error

NANOPORE: CLONING-FREE AMPLICONS

every read spans the 1,230-bp amplicon



Eight wild-type reads over the whole amplicon, with every mutation called in them; errors drawn at the rate the controls show, 1 base in 700, pale below Q 20

ILLUMINA: PAIRED SHORT READS

each pair is merged into one read, which then takes the same read route as nanopore



J_H4 (Ldha study): 143-bp fragment, 2 × 151-bp reads

R1 → adapter, dropped ← R2

merged

every base read twice; 95% of pairs merge

CH12F3 VDJ: 269-bp fragment, 2 × 150-bp reads

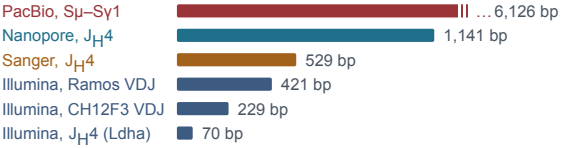
R1 → ← R2

merged

31 bp read twice, the rest once; 97% merge

read twice (Q60) read once | disagreement: better call kept, N on a tie

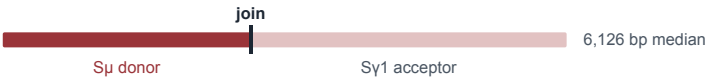
How much of the locus one read covers



Illumina settings: minimap2 sr preset; error gate 0.02 per base; minimum alignment ≤ 100 bp; then the same control test and FDR. The Ldha and CH12F3 files carry a flat Q30 (SRA), so their error model has a single quality level; the Ramos files keep binned ones.

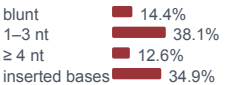
PACBIO: ONE READ ACROSS A SWITCH JUNCTION

the donor, the join and the acceptor on a single HiFi read



the join, its microhomology and any inserted bases are read, not inferred;
reads per unique junction run 1.1–23.3, so counts track PCR recovery, not switching

Microhomology at the join



From the Sanger clones here

Each molecule's own mutations
the mutation load of every clone, the share of clones mutated, and frequency as published: unique mutations over clones × a fixed length

Bookkeeping checks
copied traces, clones filed under two mice, mutated clones filed as controls, mislabeled tubes (Figure S7)

Error reported, not subtracted
the AID-null clones' error rate (about 20 per 10^5 bases at Q ≥ 20 in both published studies), and how many of each animal's calls it explains

From both, and they agree

Wild type against AID-null
Sanger 38-fold; nanopore 141-fold, with controls without calls counted at one event

Substitution spectrum
r = 0.92 (95% CI 0.74–0.98), slope 0.97

Position along the window
r = 0.94 in 25-bp bins

AID hotspot use
G:C at WRC/GYW 58% and 54% (20% of sites); A:T at WA/TW 70% and 69% (50% of sites)

Not comparable: absolute density
a clone is read once; nanopore counts unique mutations among 500 reads. Compare densities within a platform, not between.

From the nanopore reads here

The whole amplicon
positions 90–1,230: 229-fold over controls; 69% of mutations in the first 450 bp, falling 7-fold to the last 250 bp (Figure 6A)

Clones and mutation trees
clones of each mouse, and their mutations ordered into trees (Figures 6 and S11)

Calls at an estimated false-discovery rate
each candidate tested against the controls, the threshold set by them as decoys; single-read mutations not called, as one background-subtracted estimate